Critical-tail transitivity in the exceptional paired branch

Human Compression project: L044 continuation

Retain the full critical six-point-head hypotheses, including actual ambient IP, all maximal omissions intransitive and the smaller-degree alternating induction. In the contrary rank range the accepted critical-head theorem gives $n = 2r \geq 18$ and

$$\begin{aligned} &D = \{1, \dots, 6\}, \quad E = \Omega \setminus D, \\ &\tau = \rho_3 = (6\ b)\delta, \quad \sigma = \rho_4, \quad H = G_{\geq 5}, \quad K = G_{>5}, \\ &R = \langle \tau, \sigma, H \rangle = \mathrm{Alt}(E \cup \{6\}), \qquad B = \langle \sigma, H \rangle. \end{aligned}$$

The head on D is actual and supported; σ fixes D; H fixes $D \cup \{b\}$ and centralizes δ. The odd involution δ fixes $D \cup \{b\}$. No transitivity of B is assumed. All eligible edges refer to full ambient omissions. Every predecessor proof and complete required finite input remains an acquired premise.

Lemma 1 (The invariant-set replacement)**.** *The point $x = b^\sigma$ differs from b, is fixed by τ, and is fixed by K. If a nonempty proper B-invariant subset $S \subset E$ is also δ-invariant, the configuration is impossible.*

Proof. If σ fixed b, then R would preserve $\{6, b\}$, contrary to its transitivity. The full actual pure-overlap/IP lemma gives $\tau x = x$, using the natural R; its proof does not assume transitivity of B. Finally K fixes b and centralizes σ, so fixes x.

If $b \notin S$, the groups B and $\langle \tau \rangle$ both preserve S, a proper nonempty subset of $E \cup \{6\}$. If $b \in S$, they both preserve $S \cup \{6\}$ instead. Either contradicts transitivity of R. This is not a claim that arbitrary B-invariant sets are δ-invariant. □

Theorem 2 (Pairs assigned to actual orbits)**.** *Suppose every colour $j \geq 5$ has two ambient-eligible edges in some single actual H-orbit. Then the contrary rank range is impossible, without the earlier additional assumption that B is transitive on E.*

Proof. We use the whole accepted orbit-transport, equivariant aggregation, strict internal and localized internal proofs. Their combinatorial and group hypotheses are unchanged: G is transitive, the specified seven-point head is fixed by H, τ centralizes H and belongs to each relevant omission, $n = 2r$, and full actual IP and the natural critical R hold. Assign each colour exactly once to an orbit with its pair and use the accepted normalization. For assigned orbits O_s, put $d_s = |F_s| - |O_s|/2$, let f count H-fixed points and c its nontrivial orbits. The complete paid calculation gives

$$\sum_s d_s = f/2 - 5 \geq -3/2, \qquad f + c \leq 10.$$

It excludes odd exchanged nontrivial orbits, nonfirst internal odd orbits, and empty internal odd orbits of size at least six. The remaining nonempty first-assigned budgets are precisely $(f, c) = (8, 1)$ or $(7, 2)$, with an internal orbit of size $2q$, q odd; the latter also has the odd anchored orbit and final localized interval of the accepted proof.

Here are all changed transitivity interfaces in that argument. In the $(8, 1)$ case let z be the sole additional H-fixed point outside $D \cup \{b\}$. The involution δ fixes z because it centralizes H and preserves the other seven fixed points. If $x = z$, then $S = \{b, z\}$ is preserved by B and δ,

contradicting Lemma 1. Otherwise x lies in the free internal orbit, contrary to $\tau x = x$. In the $(7, 2)$ case, the original proof merely needs $x \neq b$, $\tau x = x$ and $Kx = x$, all supplied by Lemma 1. It places x in the odd orbit fixed pointwise by τ. The complete localized interval inequality then excludes that budget unchanged.

The remaining empty internal pair or pair of exchanged fixed points reduces, by the paid fixed-point count, to $\tau = (6\, b)(u\, v)$, with H preserving $\{u, v\}$ and transitive on its complementary nontrivial orbit. The two-point rotation/localization proof uses B-transitivity only to obtain $x \neq b$ and to show that σ cannot preserve $\{u, v\}$. The first follows from Lemma 1; for the second use the same lemma with $S = \{u, v\}$, which δ preserves. Every later step is unchanged: moving just one point produces incompatible simultaneous three-cycles in the actual dihedral intersection; moving both confines all eligible later edges to a set Y with $2|I| = |Y|$, contradicting $2|I| \leq |Y| - o_{G_I}(Y)$. Singleton orbits are still counted.

This checks every use of B-transitivity in the acquired internal case proofs. All possible odd contributions to the action of τ outside its head are therefore excluded. Its exterior action is odd, the required contradiction. □

Theorem 3 (A later unique ambient colour always exists)**.** *In every critical six-point-head configuration, some colour $j \geq 5$ has exactly one ambient-eligible edge.*

Proof. Assume each such colour has at least two eligible edges. The complete global-pair proof uses the actual orbit bound $f + c \leq 10$, $f \geq 7$, and the odd central action of τ. We retain that full proof and its selected-localization and maximal-string inputs. We check the only points where its previous statement assumed B-transitivity.

If $c = 1$, or if free-orbit equivariance puts each colour's two edges in a single orbit, apply Theorem 2. If $c = 2$ and τ exchanges the two nontrivial odd orbits, then $f = 8$ and x is the additional fixed point, by $\tau x = x$. Thus $S = \{b, x\}$ is B-invariant and fixed by δ, contradicting Lemma 1. This is the only changed step in that two-orbit case.

For $c = 3, f = 7$, the preserved free-orbit alternative uses only the exact partition of labels and two inequalities $|L| \leq q - 1$, $|F| \leq (v - 3)/2$, against $r - 5 = q + (v - 3)/2$; it is unchanged. In the exchanged-row alternative, the complete equality argument gives two odd rows O, O' exchanged by τ and an odd orbit V fixed by τ. It proves that G_F is actually supported and transitive on V, and that the row path labels form a consecutive interval; no projected IP is inferred. If the first tail label 5 lies among the row labels, then σ preserves V. The set $S = V$ is preserved by B and fixed by δ, so Lemma 1 is the contradiction. If $5 \in F$, the same paid orbit/path argument forces the row labels to be terminal and makes σ preserve $O \cup O'$. This time $S = O \cup O'$ is B-invariant and δ exchanges its rows. Lemma 1 again applies.

The acquired case partition exhausts the possible odd exterior actions. Hence not every colour has two eligible edges. Every colour has at least one, since its omission is intransitive and the whole group is transitive. The claimed unique edge follows. □

Exact reuse and source scope. This is an additive hypothesis extension of the exceptional parts of *Completion of the ambient paired-head branch* and *Global eligible pairs behind a six-point head*. Their complete proofs remain paid, together with *Strict row exits and critical internal orbit budgets* and *Localized eligible edges and the final internal orbit cases*. The displayed invariant-set replacements identify each changed inference; none asserts that the old stronger hypothesis follows. The source route is the later-crossing existence step in Propositions 5.31–5.32, physical pages 38–45. The argument does not identify the resulting cut as unshared, settle the remaining opposite ends, or recover all of L044. No new computation, formalization, original or question credit is claimed.